

PH-214 Modern Physics Notes

Useful Math

Taylor series expansions:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$

$$(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n = 1 + mx + \frac{m(m-1)}{2} x^2 + \frac{m(m-1)(m-2)}{6} x^3 + \dots$$

Note: For small x , higher order terms reduce to zero

Euler's Identity:

Use complex exponentials to manipulate complicated trig functions and complex amplitudes.

$$e^{ix} = \cos x + i \sin x$$

Visualizing propagating plane waves:

A 1D plane wave propagating in the $+z$ direction can be mathematically represented as:

$$\psi = f(kz - \omega t) \iff f(z - vt), \quad v = \frac{\omega}{k}$$

Where:

- $(kz - \omega t)$ [rad]:

The argument represents the **phase** of the wave. A change in phase simply moves the wave forward or backward in space and time, without changing its shape, there are a few ways to see this which we will look at later.

- k [rad/m]:

is the **wave number**, representing the spatial frequency of the wave, or how *spread out* the wave is in space. Since k scales the spatial variable z , a larger k means the wave advances through the same phase over a smaller change in z .

- ω [rad/s]:

is the **angular frequency**, representing the temporal frequency of the wave, or how *spread out* the wave is in time. As in rotational kinematics, ω scales time t , and a larger ω means the wave advances through the same phase over a smaller change in t .

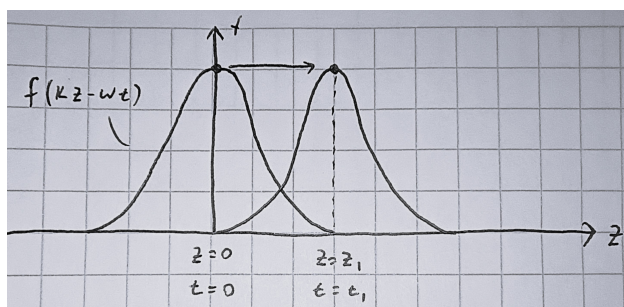
A plane wave is special in that it represents a wave whose shape is fixed as it propagates through space and time, with its shape represented by the function f .

Note:

$$k = \frac{2\pi}{\lambda}, \quad \frac{\omega}{2\pi} = f = \frac{1}{T} [\text{Hz}], \quad v = \frac{\omega}{k}$$

Global View

As mentioned before, there are two ways to visualize how the phase ($kz - \omega t$) represents a propagating wave. The first is a *global* view, looking at how z and t change relative to each other for a fixed phase, tracking a specific point on the wave.



When we track a specific point on the wave, f must give the same value at different positions and times, meaning the phase must stay constant. This shows that as t increases, z must also increase to maintain the phase (the wave moves rightward!). Vice versa, as z increases, t must increase too to keep the phase constant (again, moving rightward!).

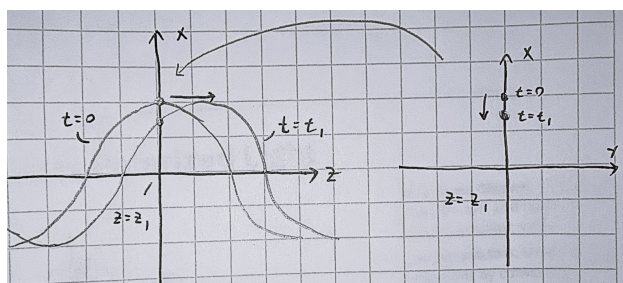
Note: This intuition applies to *any* point on the wave! Thus, by tracking all points, we can clearly see a propagating plane wave.

Local View

The second is a *local* view, looking at how the phase changes at a fixed position as time changes, or at a fixed time as position changes. For this, we will look at a sinusoidal wave, to better show oscillation: $\psi = \cos(kz - \omega t)$

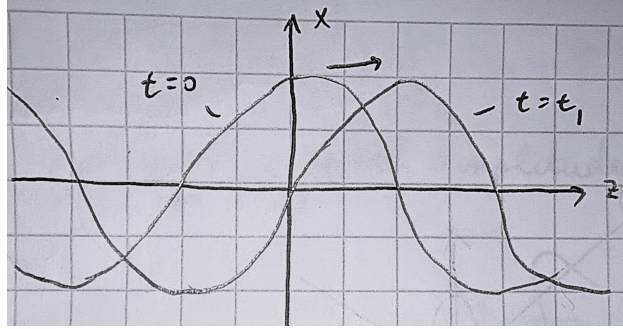
1. Fixing z , letting t vary:

Imagine we are at a fixed position $z = z_1$. As time increases, we are effectively watching a cross-section of the wave at that point, seeing only a single point along the wave moving up and down like riding the “surface” of the wave.



2. Fixing t , letting z vary:

Now imagine we are at a fixed time $t = t_1$. Moving along the z axis, and plotting the wave's amplitude, we see the full shape of the wave at that instant (ωt_1 becomes a constant phase shift).



3. Letting z and t vary together:

When we looked at z and t separately, we saw only a *part* of the wave's behavior. For a fixed z , we observed how a single point moves over time, capturing the wave's temporal variation. For a fixed t , we saw the spatial variation, revealing the full shape of the wave statically at that instant. When both z and t vary together, we can see the wave's complete dynamic behavior, watching how its shape propagates through space and time.

Note: This intuition will become very useful when we study electromagnetic polarization, where the plane wave is a 2D wave. In our cross-sectional view, the single point along the wave will move along the xy plane instead of just up and down.

Vector Calculus

- **Del Operator:**

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

- **Laplacian Operator:**

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad (\text{in cartesian})$$

- **Divergence:** Dot product of del operator with a vector field $\vec{A}$, giving a scalar field that represents how much $\vec{A}$ spreads (diverges) out from a point.

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Note: A positive divergence indicates a source, while a negative divergence indicates a sink.

- **Curl:** Cross product of del operator with a vector field $\vec{A}$, giving a vector field that represents how much $\vec{A}$ curls around a point.

$$\begin{aligned}\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \underbrace{\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right)}_{\text{curl about } x\text{-axis}} \hat{x} + \underbrace{\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right)}_{\text{curl about } y\text{-axis}} \hat{y} + \underbrace{\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)}_{\text{curl about } z\text{-axis}} \hat{z}\end{aligned}$$

Note: A positive curl indicates a counterclockwise rotation, while a negative curl indicates a clockwise rotation about the respective axis.

- **Vector Identities:**

Triple Products

$$\begin{aligned}\vec{A} \cdot (\vec{B} \times \vec{C}) &= \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) \\ \vec{A} \times (\vec{B} \times \vec{C}) &= \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})\end{aligned}$$

Product Rules

$$\begin{aligned}\vec{\nabla}(fg) &= f(\vec{\nabla}g) + g(\vec{\nabla}f) \\ \vec{\nabla}(\vec{A} \cdot \vec{B}) &= \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A}) + (\vec{A} \cdot \vec{\nabla})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} \\ \vec{\nabla} \cdot (f\vec{A}) &= f(\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla}f) \\ \vec{\nabla} \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \\ \vec{\nabla} \times (f\vec{A}) &= f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla}f) \\ \vec{\nabla} \times (\vec{A} \times \vec{B}) &= (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{A}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{A})\end{aligned}$$

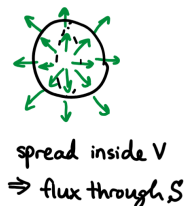
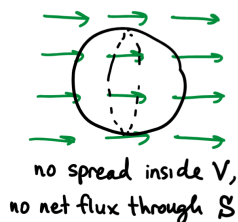
Second Derivatives

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) &= 0 \quad (\text{Divergence of a curl is always zero}) \\ \vec{\nabla} \times (\vec{\nabla}f) &= 0 \quad (\text{Curl of a gradient is always zero}) \\ \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) &= \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad (\text{Curl of a curl})\end{aligned}$$

Divergence Theorem

Relates the flux of a vector field $\vec{F}$ through a closed surface S to the divergence of $\vec{F}$ within the volume V enclosed by S .

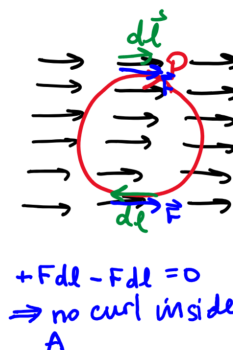
$$\underbrace{\int_V (\vec{\nabla} \cdot \vec{F}) d\tau}_{\text{how field spreads inside the volume}} = \underbrace{\oint_S \vec{F} \cdot d\vec{a}}_{\text{flux of field through the surface}}$$



Stokes Theorem

States that inside a surface A bounded by a closed curve P , if a vector field $\vec{F}$ has a non-zero curl about $\vec{A}$ (the normal to the surface), then there will be a non-zero circulation of $\vec{F}$ around the curve P .

$$\underbrace{\int_A (\vec{\nabla} \times \vec{F}) \cdot d\vec{a}}_{\text{flux of the curl of } \vec{F}} = \underbrace{\oint_P \vec{F} \cdot d\vec{l}}_{\text{circulation of } \vec{F}}$$



Key Electricity and Magnetism Laws

- Coulomb's Law

$$\vec{F}_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

- Magnetic Force

$$\vec{F}_b = q\vec{v} \times \vec{B}$$

- Electric Field of a Point Charge
sets up a diverging field

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

- Biot-Savart Law
sets up a curling field

$$\vec{B} = \int \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2}$$

Unit 1: Electromagnetic Waves

Maxwell's Equations

1. Gauss's Law for Electricity

Relates the electric flux through a closed 3D Gaussian surface to the total charge enclosed within that surface.

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0}$$

There exists a diverging electric field if there exists a non-zero charge density at that point.

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

2. Gauss's Law for Magnetism

Any 3D Gaussian surface will have zero net magnetic flux (no magnetic monopoles).

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

There exists no diverging magnetic field at any point in space because there are no magnetic monopoles.

$$\vec{\nabla} \cdot \vec{B} = 0$$

3. Faraday's Law of Induction

Relates the electric circulation around a closed Faradian loop to the changing magnetic flux through the surface bounded by the loop.

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Note: For coils with N turns, multiply the flux by N .

A changing magnetic field induces a circulating (curling) electric field.

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. Ampere-Maxwell Law

Relates the magnetic circulation around a closed Amperian loop to the enclosed current and changing electric flux through the surface bounded by the loop.

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

Both a changing electric field and a non-zero current density induce a circulating (curling) magnetic field.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}, \quad I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}$$

Deriving the Differential Forms of Maxwell's Equations

By applying the divergence and stokes theorems to the integral forms of Maxwell's equations, we can derive their corresponding differential forms. This process allows us to express the behavior of electric and magnetic fields locally at every point in space and time, rather than just over closed surfaces or loops.

1. Gauss's Law for Electricity

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0}$$

Applying the divergence theorem to the left-hand side:

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{E}) d\tau$$

We can also write the enclosed charge as a volume integral of the charge density:

$$q_{\text{enc}} = \int_V \rho d\tau \quad \rho = (\text{vol charge density})$$

Thus,

$$\begin{aligned} \implies \int_V (\vec{\nabla} \cdot \vec{E}) d\tau &= \frac{1}{\epsilon_0} \int_V \rho d\tau \\ \implies \boxed{\vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\epsilon_0}} \end{aligned}$$

2. Gauss's Law for Magnetism

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

Applying the divergence theorem:

$$\oint_S \vec{B} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{B}) d\tau = 0$$

Thus,

$$\implies \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

3. Faraday's Law of Induction

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a}$$

We can also write the changing magnetic flux as:

$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt} \left[\int_S \vec{B} \cdot d\vec{a} \right] = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

Note: Stokes theorem relates the circulation around a curve to the curl over a surface at a single instant in time. In other words, the surface S is *fixed* and does not change in time, so $d\vec{a}$ is *time independent* and the time derivative acts only on the integrand, allowing us to move the time derivative inside the integral.

Thus, equating the two sides gives:

$$\begin{aligned} \Rightarrow \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} &= - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a} \\ \Rightarrow \boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \end{aligned}$$

4. Ampere-Maxwell Law

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{B} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a}$$

We can also write the enclosed current as a surface integral of the current density:

$$I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}, \quad \vec{J} = (\text{area current density})$$

And the changing electric flux as:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt} \left[\int_S \vec{E} \cdot d\vec{a} \right] = \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a}$$

Thus, equating the two sides gives:

$$\begin{aligned} \Rightarrow \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} &= \int_S \left(\mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{a} \\ \Rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}} \end{aligned}$$

Applying Vector Identities

Additionally, we can apply the vector identities we have to the differential forms of Maxwell's equations to derive some useful results.

Applying divergence of a curl is zero to Faraday's law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) &= 0 \\ \vec{\nabla} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) &= 0 \\ -\frac{\partial}{\partial t} (\underbrace{\nabla \cdot \vec{B}}_{\text{Gauss's law for magnetism (=0)}}) &= 0 \\ \implies 0 &= 0\end{aligned}$$

Applying divergence of a curl is zero to Ampere-Maxwell law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) &= 0 \\ \vec{\nabla} \cdot \left(\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\underbrace{\nabla \cdot \vec{E}}_{\text{Gauss's law for electricity } (= \frac{\rho}{\epsilon_0})}) &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\epsilon_0} \right) &= 0 \\ \implies \boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} &\quad (\text{Continuity Equation})\end{aligned}$$

We can rewrite the continuity equation into an integral form:

Integrating both sides over a volume V :

$$\int_V (\vec{\nabla} \cdot \vec{J}) d\tau = \int_V -\frac{\partial \rho}{\partial t} d\tau$$

Apply divergence theorem to the left-hand side:

$$\int_V (\vec{\nabla} \cdot \vec{J}) d\tau = \oint_S \vec{J} \cdot d\vec{a}$$

Since the volume V is fixed in space, we can pull the time derivative outside the integral.

We also know that the total charge enclosed is $q = \int_V \rho d\tau$:

$$\int_V -\frac{\partial \rho}{\partial t} d\tau = -\frac{d}{dt} \underbrace{\int_V \rho d\tau}_{q_{\text{enclosed}}} = -\frac{dq}{dt}$$

Thus, equating the two sides gives:

$$\oint_S \vec{J} \cdot d\vec{a} = -\frac{dq}{dt}$$

Note: The continuity equation is also called a flux based conservation law. It represents a local **conservation of charge** and states that if the total charge enclosed in a volume is changing ($\frac{dq}{dt} \neq 0$), then there must be a net current flowing into or out of the volume ($\oint_S \vec{J} \cdot d\vec{a} \neq 0$) that accounts for that change.



If a balloon were filled with a certain amount of water and that quantity of water was decreasing...



It must mean that there is a flux of water, passing through the surface, leaving the volume!

Showing that E and B Fields Exist as Waves in a Vacuum

In a vacuum, there are no charges or currents, so we can set $\rho = 0$ and $\vec{J} = \vec{0}$ in Maxwell's equations. This simplifies them to:

$\vec{\nabla} \cdot \vec{E} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	no divergent fields (no source charge)
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$	only exist curling fields induced by one another!

By applying the curl of a curl rule to Faraday's and Ampere-Maxwell laws, we can derive the wave equations for the electric and magnetic fields in a vacuum.

$\begin{aligned} \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) &= \underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{E})}_{0 \text{ in a vacuum}} - \nabla^2 \vec{E} \\ \vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t} \right) &= -\nabla^2 \vec{E} \\ -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B}) &= -\nabla^2 \vec{E} \\ -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= -\nabla^2 \vec{E} \end{aligned}$	$\begin{aligned} \vec{\nabla} \times (\vec{\nabla} \times \vec{B}) &= \underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{B})}_0 - \nabla^2 \vec{B} \\ \mu_0 \epsilon_0 \frac{\partial}{\partial t} \underbrace{(\vec{\nabla} \times \vec{E})}_{-\frac{\partial \vec{B}}{\partial t}} &= -\nabla^2 \vec{B} \\ -\mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} &= -\nabla^2 \vec{B} \end{aligned}$
$\Rightarrow \boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$	$\Rightarrow \boxed{\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}}$

Recall: The classic wave equation is

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \Rightarrow \quad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

Modeling EM Radiation (Light) in a Vacuum

Properties of light:

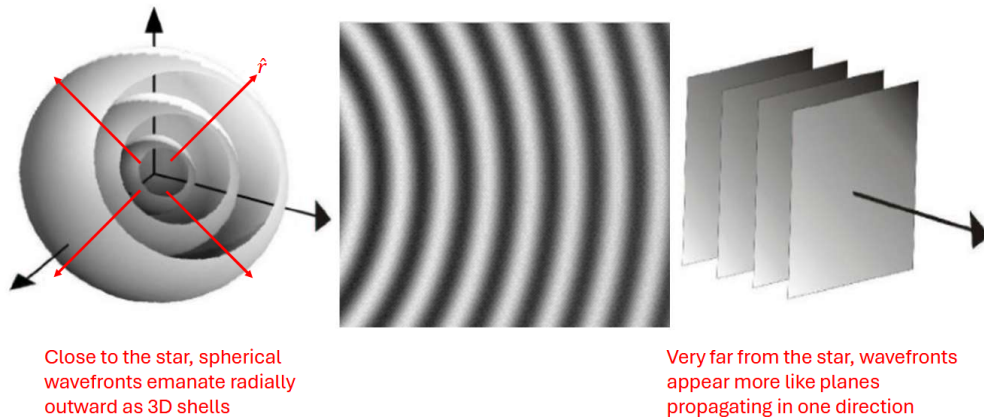
- $\vec{E} \times \vec{B}$ points in the direction of propagation.
- $\vec{E}$ and $\vec{B}$ are perpendicular to each other and to the direction of propagation (transverse wave).
- $\vec{E}$ and $\vec{B}$ oscillate in phase as to perpetually induce each other according to Faraday's and Ampere-Maxwell laws.
- The magnitudes of $\vec{E}$ and $\vec{B}$ are related by $E = cB$.
- Light does not require a medium to propagate.
- Light has speed

$$v = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Our goal is to find a solution for $\vec{E}$ and $\vec{B}$ that satisfies the 3D wave equations as well as Maxwell's equations in a vacuum.

Plane Wave Approximation

Consider the following wavefronts generated by a point source of light (ie. a star):

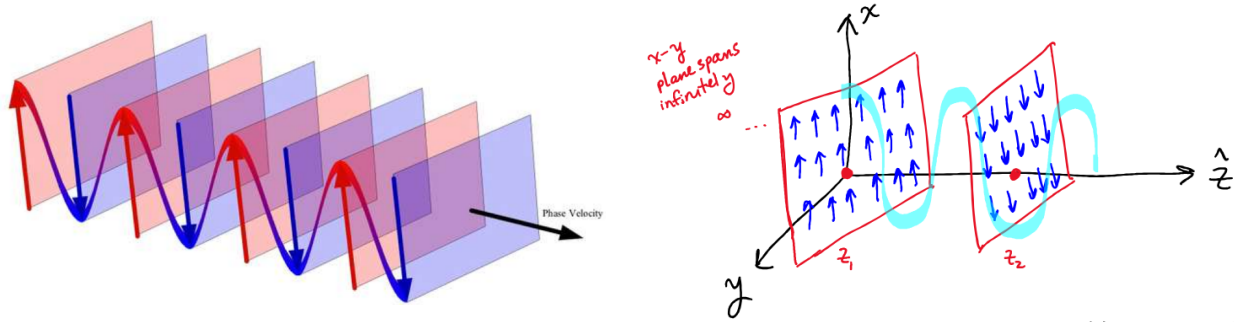


The radius of curvature of the wavefronts increases as they travel outward. Thus, at a far enough distance from the source, the wavefronts will appear to be flat planes consisting of electric and magnetic fields with *uniform* magnitude and direction. This is called the **plane wave approximation**.

The plane wave model also assumes **monochromatic light**, having a fixed wavelength λ and wavenumber k ($k = 2\pi/\lambda$). This also implies a fixed frequency ω , since the speed of light is constant:

$$v = \frac{\omega}{k} = c \quad \implies \quad \omega = ck.$$

Without a single frequency, the wavefronts would interfere with one another and would no longer form the ideal uniform planes described by the plane wave approximation.

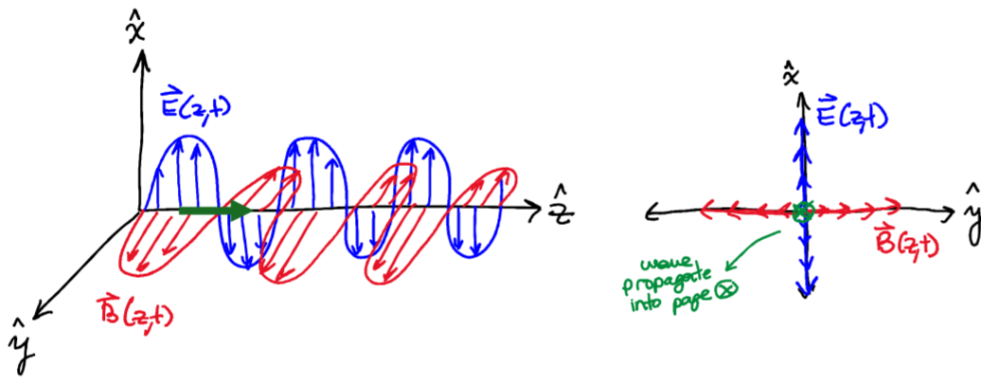


Note: Although light consists of both electric and magnetic fields, we will primarily focus on the electric field when modeling light, since the magnetic field can be readily derived from it using Faraday's law. We will also focus on the electric field when analyzing polarization, since the direction of the electric field determines the polarization of light.

Case $\hat{z}$

For the case $\hat{z}$, we take the direction of propagation to be along the $+z$ axis. At any fixed position z , the electric field $\vec{E}$ has a uniform magnitude and direction across the entire xy plane. As with the 1D plane wave, the *local view* at a fixed point along the z axis shows the electric field oscillating in time as the light wave passes through, with the electric field remaining transverse to the direction of propagation.

Note: Don't forget that the magnetic field $\vec{B}$ is also oscillating in phase with and is perpendicular to both $\vec{E}$ and the direction of propagation.



Therefore, a solution for the electric field of a plane wave in case $\hat{z}$ is:

$$\vec{E} = \underbrace{(E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y})}_{\text{complex amplitude vector}} e^{i(kz - \omega t)}$$

From the equation, we can see that the magnitudes of the electric field in the x and y directions need not be the same, and that there is no z component of the electric field, since the wave is transverse. Like the 1D plane wave, the space and time dependence of this 3D plane wave is contained in the phase $(kz - \omega t)$.

The amplitude of the electric field is given by

$$|\vec{E}| = \sqrt{|E_{0x}|^2 + |E_{0y}|^2}$$

Note: The phases ϕ_x and ϕ_y of the complex amplitude vector do not affect the magnitude of the electric field, but they do govern the polarization of the light wave. This is because of the dot product:

$$\begin{aligned} |\vec{E}|^2 &= \vec{E} \cdot \vec{E}^*, \quad \vec{E}^* = (\text{complex conjugate of } \vec{E}) \\ \vec{E} \cdot \vec{E}^* &= (E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) \cdot (E_{0x} e^{-i\phi_x} \hat{x} + E_{0y} e^{-i\phi_y} \hat{y}) = |E_{0x}|^2 + |E_{0y}|^2 \end{aligned}$$

The exponentials multiply to 1.

Note: The xy components for both the electric and magnetic fields have the same harmonic properties of k and ω (wavelength and frequency), since they are both derived from the same phase $(kz - \omega t)$.

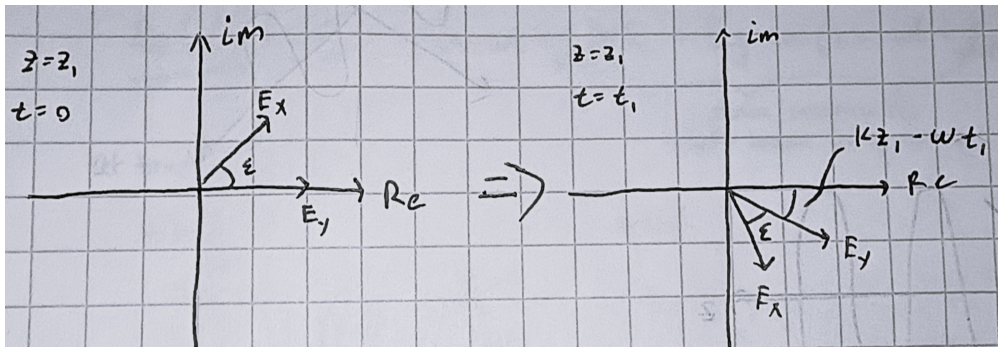
Finally, note that we have switched from sinusoidal functions to complex exponentials. This is because complex exponentials are easier to work with mathematically, and if we want to get back to the real-valued (physically measurable) electric field, we can simply take the real part of the complex expression (using Euler's formula).

$$\begin{aligned} \vec{E}_{\text{real}} &= \Re\{\vec{E}\} = \Re\{(E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) e^{i(kz - \omega t)}\} \\ \vec{E}_{\text{real}} &= E_{0x} \cos(kz - \omega t + \phi_x) \hat{x} + E_{0y} \cos(kz - \omega t + \phi_y) \hat{y} \end{aligned}$$

Polarization of Light

Polarization has been mentioned a few times already, but what exactly is it? Simply put, polarization describes the path traced out by the tip of the electric field vector $\vec{E}$ in the xy plane at a fixed z as the light wave passes through it. How can we visualize this? With *parametric equations*.

First, to better understand the phase difference between the x and y components of the electric field, we draw a phasor diagram in the complex plane:



Where

$$E_x = E_{0x} e^{i(\phi_x + kz_1 - \omega t)}$$

$$E_y = E_{0y} e^{i(\phi_y + kz_1 - \omega t)}$$

$$\boxed{\varepsilon = \phi_x - \phi_y \quad (\text{phase difference})}$$

Since $z = z_1$ is fixed, kz_1 becomes a constant phase offset, so the rotation of the phasors is driven entirely by $-\omega t$ (CW, since it is negative). The phase $kz_1 - \omega t$ is the same for both E_x and E_y , so the relative phase difference ε between the two components remains constant as the light wave propagates. Also note that while the magnitudes of E_x and E_y are constant in the phasor diagram, their *real* magnitudes are given by the projections of the phasors onto the real axis, which oscillate in time (allowing elliptical polarization, where the magnitude of $\vec{E}$ varies in time).

General Polarization

The most general case of polarization is elliptical polarization, where linear and circular polarization are just degenerate ellipses. To see this, we will rewrite the *real* electric field components in terms of the phase difference ε :

$$\begin{aligned} \vec{E}_{\text{real}} &= E_{0x} \cos(kz_1 - \omega t + \phi_x) \hat{x} + E_{0y} \cos(kz_1 - \omega t + \phi_y) \hat{y} \\ x(t) &= E_{0x} \cos(\omega t - \varepsilon) \\ y(t) &= E_{0y} \cos(\omega t) \end{aligned}$$

Note: kz_1 is constant, so it cancels out when we take the difference of the two phases, but we keep ωt in the argument since it is the time-varying part. Because we define ε to be positive, it appears in the x component (which is why E_x leads E_y in the phasor diagram). The argument of the cosine is $\varepsilon - \omega t$, but since cosine is an even function, we can equivalently write it as $\omega t - \varepsilon$.

We now deparametrize the equations to obtain a single equation in x and y that describes the path traced out by $\vec{E}$ in the xy plane. First expanding the cosine using sum and difference:

$$\begin{aligned} x(t) &= E_{0x} (\cos(\omega t) \cos(\varepsilon) + \sin(\omega t) \sin(\varepsilon)) \\ y(t) &= E_{0y} \cos(\omega t) \implies \cos(\omega t) = \frac{y}{E_{0y}} \end{aligned}$$

Substituting $\cos(\omega t)$ into the x equation:

$$x(t) = E_{0x} \left(\frac{y}{E_{0y}} \cos(\varepsilon) + \sin(\omega t) \sin(\varepsilon) \right)$$

Solving for $\sin(\omega t)$:

$$\sin(\omega t) = \frac{1}{\sin(\varepsilon)} \left(\frac{x}{E_{0x}} - \frac{y}{E_{0y}} \cos(\varepsilon) \right)$$

Plugging $\sin(\omega t)$ and $\cos(\omega t)$ into the Pythagorean identity $\cos^2(\omega t) + \sin^2(\omega t) = 1$:

$$\left(\frac{y}{E_{0y}} \right)^2 + \left(\frac{1}{\sin(\varepsilon)} \left(\frac{x}{E_{0x}} - \frac{y}{E_{0y}} \cos(\varepsilon) \right) \right)^2 = 1$$

Simplify:

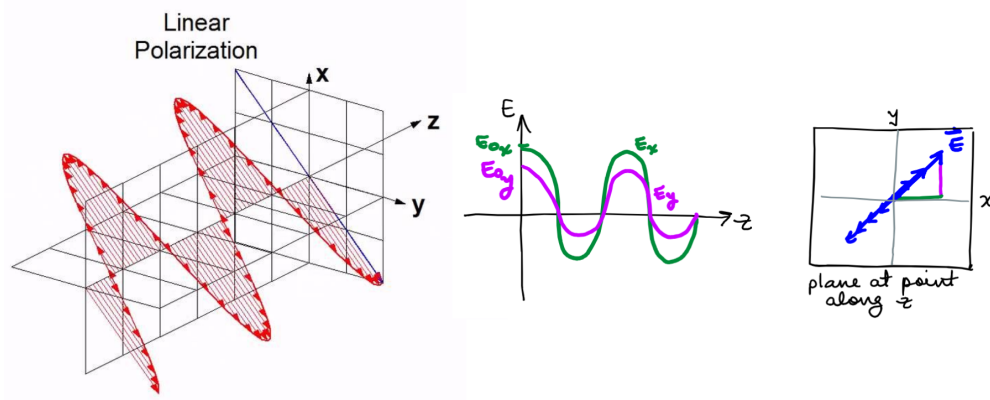
$$\frac{x^2}{E_{0x}^2} + \frac{y^2}{E_{0y}^2} - 2\frac{xy}{E_{0x}E_{0y}}\cos(\varepsilon) = \sin^2(\varepsilon)$$

Note: This is the form of an ellipse centered at the origin ($ax^2 + bxy + cy^2 = d$)

Therefore, for any ε that doesn't result in a degenerate case, the polarization of light is elliptical. Now we will list some cases so we don't have to keep referring back to the general equation:

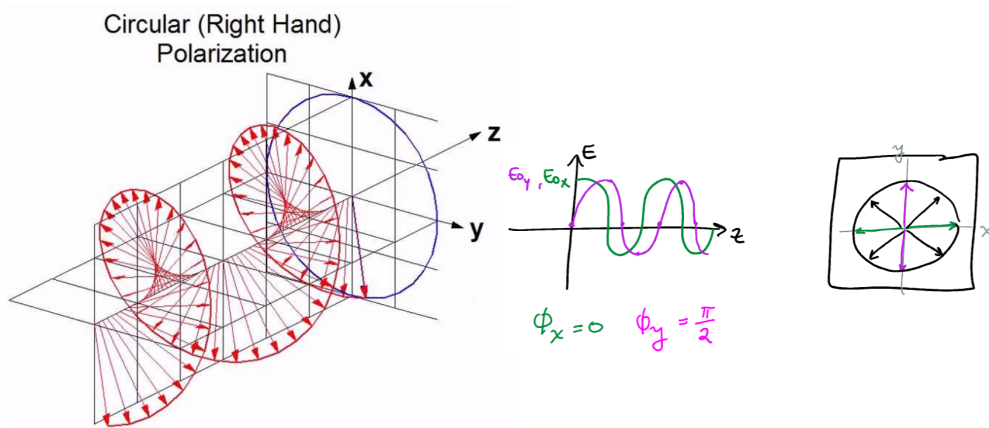
- **Linear Polarization:** E_x and E_y are in phase and the ratio of their magnitudes is constant (slope of the line).

$$\varepsilon = n\pi, \quad n = (0, 1, 2, \dots)$$



- **Circular Polarization:** E_x and E_y have equal magnitudes and are $\pi/2$ out of phase.

$$E_{0x} = E_{0y}, \quad \text{and} \quad \varepsilon = (2n + 1)\frac{\pi}{2}, \quad n = (0, 1, 2, \dots)$$

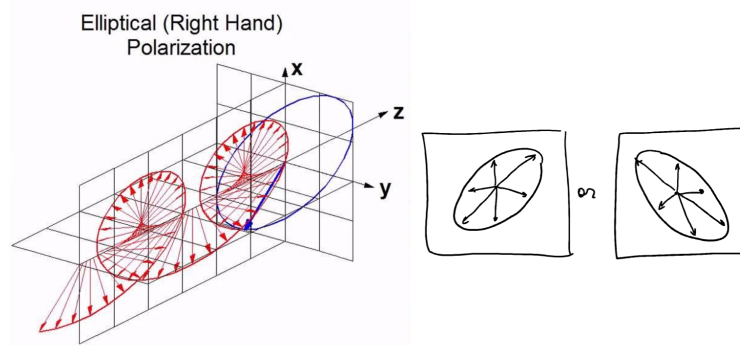


- **Elliptical Polarization:** All other cases

$$E_{0x} \neq E_{0y}, \quad \text{and} \quad \varepsilon \neq n\pi \quad (\text{otherwise it will be linear})$$

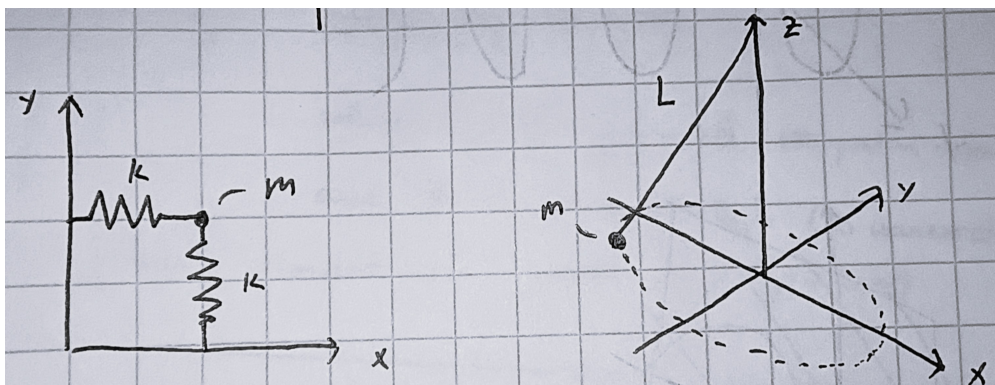
OR

$$E_{0x} = E_{0y}, \quad \text{and} \quad \varepsilon \neq (2n+1)\frac{\pi}{2} \quad (\text{otherwise it will be circular})$$



Why Ellipses?

Fundamentally, the electric and magnetic fields behave like oscillators, so the path traced out by the electric field vector must be periodic and closed. The only conic section that is both periodic and closed is an ellipse, with circles and lines appearing as degenerate cases. Another way to visualize this is to think of E_x and E_y as two simple harmonic oscillators (SHOs) that are coupled through a common driving frequency, analogous to two springs with equal spring constant k attached to the same mass. The motion of the mass is then a superposition of the two SHOs, and the resulting path traced out by the mass is an ellipse. Feynman also introduces the picture of a string-mass pendulum, where the mass can swing in any direction in the xy plane, and the path traced out by the mass is likewise an ellipse. This intuition is useful because it emphasizes that the driving phase ωt is the same for both the x and y components, and that the phase difference, and thus the resulting trace, depends only on the initial orientation of the motion.



General Plane Wave Solution

We would like to generalize the case $\hat{z}$ solution

$$\vec{E} = (E_{0x}e^{i\phi_x}\hat{x} + E_{0y}e^{i\phi_y}\hat{y}) e^{i(kz-\omega t)}$$

to describe a plane wave propagating in an arbitrary direction. To do this, we can simply replace the kz term in the phase with $\vec{k} \cdot \vec{r}$, where $\vec{k}$ is the **wavevector** that points in the direction of propagation and has magnitude k . This allows us to describe plane waves propagating in any direction, not just along the z axis.

$$\vec{k} = k_x\hat{x} + k_y\hat{y} + k_z\hat{z}$$

$$|\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = k, \quad k = \frac{2\pi}{\lambda}$$

and $\vec{r}$ is the position vector from to origin to any point in space:

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

Therefore, the general plane wave solution is:

$$\boxed{\vec{E} = (E_{0x}e^{i\phi_x}\hat{x} + E_{0y}e^{i\phi_y}\hat{y}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}}$$

This general form of the plane wave solution is intuitive because it is consistent with our understanding that $\vec{E}$ depends only on spatial variations along the direction of propagation. The dot product $\vec{k} \cdot \vec{r}$ extracts the component of $\vec{r}$ in the direction of $\vec{k}$, which for case $\hat{z}$ reduces to variations along the z axis. At the same time, $\vec{E}$ does not depend on spatial variations perpendicular to the direction of propagation (it is a transverse wave). For any $\vec{r}$ perpendicular to $\vec{k}$, we have $\vec{k} \cdot \vec{r} = 0$, so in case $\hat{z}$ there are no variations in the x and y directions.

Useful Properties of the General Plane Wave Form

Time derivative

$$\frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \quad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

Spatial derivative

$$\vec{\nabla} \cdot \vec{E} = i\vec{k} \cdot \vec{E}, \quad \nabla^2 \vec{E} = -k^2 \vec{E}$$

$$\vec{\nabla} \times \vec{E} = i\vec{k} \times \vec{E}$$

Note: These properties follow from the fact that the only spatial and temporal dependence of $\vec{E}$ is contained in the phase $(\vec{k} \cdot \vec{r} - \omega t)$. When taking derivatives, the complex amplitude vector can therefore be treated as a constant, and the chain rule applies directly to the exponential term.